

GRIDBENCH · DESIGN DECISION RECORD

GridBench TRL 2 review decisions

Open Hardware Portfolio · amishchadha.com

DOCUMENT CONTROL

Document ID	GBN-DDR-001	Version	0.1
Status	Draft	Date	2026-09-25
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/gridbench		
Source	docs/decisions/0001-trl2-review-decisions.md @ 91bc975		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-09-25	Amish Chadha	Record the TRL 2 review items adopted as recommended for TRL 3 work, and the items that remain open

0001: TRL 2 review decisions

- **Date:** 2026-09-25
- **Status:** proposed. The recommendations for items A1 to A10 are adopted for TRL 3 work pending Amish's review; items O1 and O2 remain "Proposed, awaiting Amish".

■ Context

The TRL 2 review note (docs/REVIEW.md , session 2026-09-25, /populate) listed nine items as "Proposed, awaiting Amish", and the design precis GBN-PRC-001 v0.2 listed six key design choices, each with options and a recommendation. On 2026-09-25 Amish asked for this batch of repos to be taken through the usual process with the instruction "you know the drill, nothing gets past TRL 3". He has not reviewed this repo's items one by one. Every item that carries a recommendation is therefore adopted as recommended for TRL 3 under that instruction and stays open for his review. Items without a recommendation stay open, and no new budget figure is applied.

■ Options considered

The options for each item are those listed in docs/REVIEW.md (session 2026-09-25, /populate) and in GBN-PRC-001 v0.2, Key design choices and Cost.

■ Decision

Table 1. Items adopted as recommended.

#	Item	Adopted recommendation	Status
A1	Budget and cost options	Options (a) and (b): the dial indicator is user-supplied and outside the parts budget, and the tile is bought as offcut stock. R10 is redefined to cover the bench, grid, tile and fixture set without the indicator. <code>budget_usd</code> stays at \$220; option (c), raising it to \$250, is the fallback and stays open (O2).	Adopted as recommended for TRL 3 under Amish's 2026-09-25 instruction, open for his review
A2	Grid standard	25 mm pitch, M6, compatible with metric optical breadboards; a heavy 50 mm variant to be considered later	Adopted as recommended for TRL 3 under Amish's 2026-09-25 instruction, open for his review
A3	Worktop zones	Two zones: a plywood field plus a 300 x 300 mm aluminum precision tile on the same grid	Adopted as recommended for TRL 3 under Amish's 2026-09-25 instruction, open for his review
A4	Insert spacing	M6 inserts on the 50 mm sub-grid; plain 6.6 mm holes between them for pins	Adopted as recommended for TRL 3 under Amish's 2026-09-25 instruction, open for his review
A5	Frame	Bolted timber for the prototype; welded steel as a documented variant	Adopted as recommended for TRL 3 under Amish's 2026-09-25 instruction, open for his review
A6	Fixture bodies	Printed PETG with metal dowels, screws and tile; a machined option for heavily used jigs	Adopted as recommended for TRL 3 under Amish's 2026-09-25 instruction, open for his review
A7	Stability fix	Ballast on the shelf plus an optional wall anchor bracket	Adopted as recommended for TRL 3 under Amish's 2026-09-25 instruction, open for his review
A8	Fixture file convention	A short fixture metadata block in each build123d file giving the grid footprint and datum	Adopted as recommended for TRL 3 under Amish's 2026-09-25 instruction, open for his review
A9	Fixture library license	Keep CERN-OHL-S-2.0, as for all hardware in the portfolio	Adopted as recommended for TRL 3 under Amish's 2026-09-25 instruction, open for his review
A10	Tile position	At the right of the worktop, leaving the left 800 mm free for large parts	Adopted as recommended for TRL 3 under Amish's 2026-09-25 instruction, open for his review

Table 2. Items that remain open.

#	Item	Status
O1	First workshops and regions for co-design. No recommendation was made and no preference is stated, so none is chosen here.	Proposed, awaiting Amish
O2	Raise <code>budget_usd</code> from \$220 to \$250 (cost option c). Recommended at TRL 2 only as the fallback if offcut prices do not hold. GBN-CAL-001 shows core parts at \$239.20, so the fallback is now needed for R10 to be met. <code>project.yaml</code> is unchanged.	Proposed, awaiting Amish

■ Consequences

- `project.yaml` : unchanged apart from the TRL fields. `budget_usd` stays at \$220. No pitch or problem rewording was recommended, so none was applied.
- GBN-PRB-001, GBN-PRC-001 and GBN-REQ-001 are revised to v0.3. The key design choices in the precis are no longer "proposed". R10 now excludes the user-supplied dial indicator (A1). R12 is assessed with the ballast in place (A7).
- The TRL 3 calculations (GBN-CAL-001) led to four detail changes within these choices: a deeper printed toe clamp with a rated torque of 1.6 N·m, one round and one diamond locating pin per precision fixture, the fence printed in two 190 mm segments, and two concrete slabs as the ballast. With them the core parts cost \$239.20.
- GBN-CAL-001 shows R10 not met at \$220 (met at the \$250 fallback, O2) and R7 not met on the plywood field because of insert pull-out. Options for R7 are proposed in `docs/REVIEW.md` and await Amish; this record does not decide them.
- TRL 4 is on hold by Amish's instruction. Nothing in this record authorizes building or testing.